Two Heads are Better than One: A Network Attack Detection Model Based on Multimodal and Multimedia Retrieval: Supplementary Material

Ziang Li
liziang@iie.ac.cn
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China

Zhou Zhou
zhouzhou@iie.ac.cn
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China

Chengxiang Si*
sichengxiang@cert.org.cn
National Computer Network Emergency Response Technical Team/Coordination Center of China (CNCERT/CC)
Beijing, China

1 Datasets

1.1 IoT-23

This dataset is published by researchers from Czech University of Technology and contains 20 operational scenarios of infected IoT devices and 3 operational scenarios of normal IoT devices. The malicious traffic scenarios cover a wide range of malware and attack types that run on Raspberry Pi devices and utilize a variety of protocols to perform different operations, such as scanning, DoS attacks, and control communications. The normal traffic scenario records the network traffic generated by three real IoT devices under normal usage. Both the malicious and normal traffic scenarios operate in a controlled network environment but connect to the Internet without restrictions, just like real IoT devices. This paper chose a lighter version for the experimental process, with a file format of PCAP and a total size of 8.8 GB.

1.2 CIC-IDS2017

This dataset is published by researchers from the Canadian Institute for Cybersecurity at the University of New Brunswick and contains both benign and malicious scenarios. Malicious traffic scenarios cover a broad spectrum of contemporary cyber attacks—including Denial of Service (DoS), brute force, infiltration, etc, and employ various tools and protocols to simulate realistic intrusion attempts. The benign traffic scenarios, on the other hand, record normal network operations that reflect typical user behavior and standard communications in an enterprise environment. Both the malicious and benign traffic were generated in a controlled yet realistic network setting.

2 Implementation Details

In the training phase, this paper sets the maximum training epoch to 500 and the batch size to 128. the Adam optimizer [?] is used and determines the initial learning rate to be 5e-4, the weight decay to be 1e-5, the first-order momentum parameter to be 0.9, and the second-order momentum parameter to be 0.99. The parameter details of the image modal processing branch are shown in Table 1.

Where, D_i denotes the downsampling rate of the feature map of the i-th image processing block, C_{Oi} denotes the number of channels of the output feature map of the i-th image processing block, R_i denotes the spatial redundancy reduction rate of the i-th image processing block, S_i denotes the number of times that ILFEM and IMSFE are stacked in the i-th image processing block, Y is the height of the initially input feature map, and X is its width.

Table 1: The parameter details of the IMPB.

Block number	Module	Hyperparameters	Output size
1	IBEM	$D_1 = 4$ $C_{O1} = 64$	$\frac{Y}{4} \times \frac{X}{4}$
	ILFEM+ISFEM	$R_1 = 8$ $S_1 = 2$	
2	IBEM	$D_2 = 2$ $C_{O2} = 128$	$\frac{Y}{8} \times \frac{X}{8}$
	ILFEM+ISFEM	$R_2 = 4$ $S_2 = 2$	
3	IBEM	$D_3 = 2$ $C_{O3} = 320$	$\frac{Y}{16} \times \frac{X}{16}$
	ILFEM+ISFEM	$R_3 = 2$ $S_3 = 6$	
4	IBEM	$D_4 = 2$ $C_{O4} = 512$	$\frac{Y}{32} \times \frac{X}{32}$
	ILFEM+ISFEM	$R_4 = 1$ $S_4 = 2$	

In the time domain feature extraction module of the modal processing branch of the time series, the input sequence time step $T = 96$. after the patching operation, the length of each time series block is $L_b = 16$ and the step size $S_b = 8$. the dimension of the time domain feature Transformer is $D_T = 128$ and contains two Encoders. the attention mechanism with the number of heads $H = 16, d_k = 64$, the probability of dropout is 0.2, and the feedforward network with extension multiplier 4 consists of two linear layers with activation function GELU. And in the frequency domain feature extraction module, the rest of the parameters are the same as the settings in the time domain feature extraction module, except for the frequency domain Transformer potential space dimension $D_F = 512$.

*Chengxiang Si is the Corresponding author.

3 Experimental Environment and Setup

We implemented NADMR using Pytorch on a server with a Tesla V100 card. We evaluated all aspects of NADMR using evaluation metrics such as accuracy, precision, recall, macro F1 score, GPU memory usage, training time and testing time. All experiments were performed using ten-fold cross-validation, and each set of experiments was performed six times and averaged as a result.

4 Data Preprocessing

Before transforming the traffic samples to be tested into the required modal data, we anonymize the sensitive information in the traffic (e.g., IP addresses, MAC addresses, port numbers, and user identities) using the SHA-256 hash technique to avoid the model relying on this information to detect malicious traffic and to protect the user privacy at the same time. For user identity information with high confidentiality level, a random salt value is additionally introduced to ensure different hash values for the same username and password. After completing the anonymization, preprocessing is performed according to different modal requirements.

Image Modal Data Preprocessing. Slice and clean the pcap file by session and parse out the byte stream information of each packet. The byte data are spliced into a long sequence in the capture order and arranged into a two-dimensional matrix with preset widths, and the matrix elements correspond to the original byte values (0-255), which are consistent with the pixel intensities of the grayscale image. Finally the image modal data is generated by normalization.

Time series modal data preprocessing. Parsing the pcap file to extract the packet timestamps and other metadata, and grouping the packets in 100 ms time windows. Count the features (e.g., number of packets, number of bytes, protocol distribution, traffic direction, etc.) within each window to reduce the data dimensions and extract dynamic change information. Finally, the data is normalized to generate time series modal data.